

A Packing of 20 Four-Spaces in $\mathbb{R}^{12}$

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Abstract

We give 20 4-dimensional subspaces of $\mathbb{R}^{12}$ whose smallest squared chordal distance is at least 2.806619. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.721088, obtained by deleting 30 of the 50 four-spaces of the packing listed in Cohn’s current table. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 20 subspaces of dimension 4 in $\mathbb{R}^{12}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.806619 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane’s *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 12$, $n = 4$. The cell $N = 20$ is blank in Sloane’s table, in Sloane’s coordinate files and in Cohn’s table as checked on October 8, 2026. The best value implied by known packings is 2.721088, obtained by deleting 30 of the 50 four-spaces of the packing listed in Cohn’s current table. The Rankin simplex bound for this cell is $4(12 - 4)/12 \cdot 20/19 = 2.807018$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m12n4N20.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 67 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
28066194025365962137196863464875844202290683073453633880125221497646472
692577338407070215986800915327680647856060640759330999748008518352417586
9145105121708014887716064592060447613518624420322
```

denominator:

```
999999999999846348624733324780866613090044146387868419916928208195597179
967691769910009981226059975782965168461822162423375271633962271591592468
390966307181805798805775375181712323899086978227
```

B Coordinates

20 blocks of 12 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m12n4N20.txt`.

```
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